EXPERIMENT-14: EXPLORING PRO-CODE EXTENSION OF A ZOHO CREATOR APPLICATION USING ZOHO CATALYST / CUSTOM FUNCTIONS

AIM

To extend the Student Admission Management System in Zoho Creator using custom Deluge functions and, where required, Zoho Catalyst serverless services for advanced computation, external API integration, and intensive processing.

THEORY

Zoho Creator provides forms, reports, workflows, and Deluge scripting for rapid application development. More complex student admission requirements may benefit from reusable custom functions or pro-code backend services.

1. Custom Deluge Functions

Custom Deluge functions encapsulate reusable business logic inside Zoho Creator. In the Student Admission Management System, they may be used to validate student details, check required documents, apply admission rules, and call external REST APIs.

2. Zoho Catalyst Serverless Pro-Code Integration

Zoho Catalyst supports serverless backend development using languages such as Java, Node.js, and Python. If the student admission application requires processing unsuitable for a simple Creator workflow, such as processing uploaded documents, connecting to an external eligibility service, or processing large admission batches, the work can be handled by a Catalyst function and called through an authenticated API.

PROCEDURE

Step 1: Create a Custom Deluge Function

1. Open the Student Admission Management System in Zoho Creator.
2. Navigate to the application logic/functions area available in your Creator interface.
3. Choose Create Function, then define a suitable name, input arguments, and return type.
4. Write and test the Deluge logic, then save the function.
5. Associate the function with the relevant student admission workflow, such as after a record is submitted or updated.

Example Deluge Logic — Registration / Admission Validation

```
if(input.Application_ID != null)
{
    if(input.Required_Documents_Verified == true)
    {
        input.Admission_Status = "Ready for Review";
    }
    else
    {
        input.Admission_Status = "Documents Pending";
    }
}
```

Use the actual field link names from your application. If the function is called from a workflow, pass or reference the relevant field values according to the function configuration.

Step 2: Set Up a Zoho Catalyst Project (Optional)

1. Open the Zoho Catalyst Console and create a project if a separate pro-code backend is required.
2. Use the supported development environment or CLI to create a serverless function in Node.js, Java, or Python.
3. Implement the advanced task required by the student admission, such as processing uploaded documents, connecting to an external eligibility service, or processing large admission batches.
4. Deploy the function and note its API endpoint and authentication requirements.

Step 3: Connect Creator to Catalyst Using an API

1. In Creator, open the Connections area under the integration settings and create an authenticated connection if your account supports it.
2. Configure the Catalyst endpoint and authentication securely. Do not expose API keys or tokens in a public form or hardcode sensitive credentials in shared scripts.
3. Use an invokeurl request in a custom function to send only the required student admission data to the endpoint.

4. Parse the response and use the returned result in the appropriate application field or report.

Example Request Structure

```
response = invokeurl
[
    url : "YOUR_AUTHENTICATED_CATALYST_ENDPOINT"
    type : POST
    parameters : {"applicationId": input.ID, "course": input.Course_Selected}
    connection : "your_connection_link_name"
];
```

Step 4: Test and Validate the Workflow

1. Submit a test student admission record or use a safe test record.
2. Confirm that the Creator workflow triggers the custom function.
3. If Catalyst is used, verify that the request reaches the endpoint and returns the expected response.
4. Check the calculated or validated result in the student admission record and report.
5. Test invalid inputs and API failures so the application handles errors safely.

RESULT

The pro-code extension capabilities of Zoho Creator were studied using the Student Admission Management System. Reusable Deluge functions can support admission validation and document-status checks. When advanced processing is required, Zoho Catalyst can provide a separate pro-code backend through authenticated API calls. Actual implementation depends on the field link names, connections, and features available in the Zoho Creator account.